🔑 **Answer Key — Teacher Copy · Lesson 3-2**

MSTAR PRACTICE WORKSHEET

Lesson 3-2: Understand Rates and Unit Rates

Grade 6 Mathematics · Standard 6.AT.2

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

A booth charges $4.50 for 9 games. What is the unit rate?

A. $0.45 per game

B. $0.50 per game

C. $2.00 per game

D. $4.50 per game

$4.50 ÷ 9 = $0.50 per game.

Question 2 SELECTED RESPONSE — CORRECT: B

A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

A. 6 miles per gallon

B. 30 miles per gallon

C. 36 miles per gallon

D. 180 miles per gallon

180 miles ÷ 6 gallons = 30 miles per gallon.

Question 3 SELECTED RESPONSE — CORRECT: B

A pack of 8 markers costs $6.00. What is the cost per marker?

A $0.60

B $0.75

C $0.80

D $1.33

$6.00 ÷ 8 = $0.75 per marker.

Question 4 SELECTED RESPONSE — CORRECT: A

Which is a unit rate?

A 12 miles per 1 hour

B 24 miles in 2 hours

C 36 miles for 3 gallons

D 100 points in 5 games

A unit rate has 1 in the second quantity. '12 miles per 1 hour' is the only choice where the second quantity is 1.

Question 5 SELECTED RESPONSE — CORRECT: A

Token packs: Pack A is 20 tokens for $5.00, Pack B is 35 tokens for $7.00, and Pack C is 50 tokens for $12.50. Which pack is the best deal per token?

A Pack B

B Pack A

C Pack C

D They are all the same

Pack B has a unit rate of $0.20 per token, which is the lowest cost per token.

Question 6 SELECTED RESPONSE — CORRECT: A

Runner A completes 4 laps in 12 minutes. Runner B completes 5 laps in 14 minutes. Who runs more laps per minute?

(A) Runner B

(B) Runner A

(C) Same rate

(D) Cannot determine

Runner A: 4÷12 ≈ 0.33 laps/min. Runner B: 5÷14 ≈ 0.36 laps/min. Runner B runs slightly more laps per minute.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Booth A charges $3 for 5 games. What is the unit rate in dollars per game?

(A) $0.30 per game

(B) $0.60 per game

(C) $1.67 per game

(D) $15.00 per game

Unit rate = total cost ÷ number of games = 3 ÷ 5 = $0.60 per game.

- **A:** That divides 3 by 10. There are 5 games, so divide 3 by 5.
- **C:** That is 5 divided by 3, the ratio upside down. Dollars per game divides dollars BY games.
- **D:** That multiplies 3 by 5. A unit rate divides the total cost by the number of games.

Part B — correct answer: D

Booth B charges $5 for 8 games. Which booth is the better deal, and why?

(A) Booth B, because $5 buys more games than $3.

(B) Booth A, because 3 is less than 5.

(C) They are equal, because both ratios simplify to the same value.

(D) Booth A, because $0.60 per game is less than $0.625 per game.

Booth B's unit rate is 5 ÷ 8 = $0.625 per game. Comparing per-game costs, $0.60 < $0.625, so Booth A costs less per game.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Dividing the Wrong Way

> Marisol wanted the cost per pound of apples. The sign read '4 pounds for $6.' She wrote: 'Unit rate = 4 ÷ 6 = about $0.67 per pound.'

Explain Marisol's misconception and give the correct unit rate.

Model answer: 6 ÷ 4 = $1.50 per pound. The words 'per pound' say what to divide by: dollars ÷ pounds. Marisol's 4 ÷ 6 ≈ 0.67 is pounds per dollar, a different rate that answers a different question.

2 points	Student identifies that 'cost per pound' means dollars divided by pounds, computes 6 ÷ 4 = $1.50 per pound, and explains that 4 ÷ 6 gives pounds per dollar instead.
1 point	Student gives $1.50 per pound without explaining which quantity is divided by which.
0 points	Student agrees with Marisol or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

An arcade sells tokens in three packs: 25 tokens for $5, 60 tokens for $10.80, and 100 tokens for $19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Model answer: The student is not correct. Dividing each price by its tokens gives $5 ÷ 25 = $0.20 per token, $10.80 ÷ 60 = $0.18 per token, and $19 ÷ 100 = $0.19 per token. The 60-token pack has the lowest unit rate, so it is the best deal — getting the most tokens is not the same as paying the least for each one.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.